

FIITJEE

ALL INDIA TEST SERIES

FULL TEST – VIII

JEE (Main)-2022

TEST DATE: 06-03-2022

Time Allotted: 3 Hours

Maximum Marks: 300

General Instructions:

- The test consists of total 90 questions.
- Each subject (PCM) has 30 questions.
- This question paper contains **Three Parts**.
- **Part-A** is Physics, **Part-B** is Chemistry and **Part-C** is Mathematics.
- Each part has only two sections: **Section-A and Section-B**.
- **Section – A** : Attempt all questions.
- **Section – B** : Do any five questions out of 10 Questions.

Section-A (01 – 20, 31 – 50, 61 – 80) contains 60 multiple choice questions which have **only one correct answer**. Each question carries **+4 marks** for correct answer and **-1 mark** for wrong answer.

Section-B (21 – 30, 51 – 60, 81 – 90) contains 30 Numerical answer type questions with answer XXXXX.XX and each question carries **+4 marks** for correct answer. There is no negative marking.

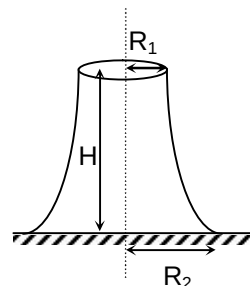
Physics

PART – A

SECTION – A (One Options Correct Type)

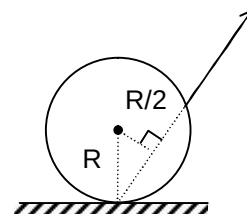
This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

1. A solid object having uniform density (ρ) and varying cross-sectional area (which is circular) is placed on smooth horizontal surface as shown. Then the value of $\frac{R_1}{R_2}$ if the stress throughout its height remains same and having value equal to c .



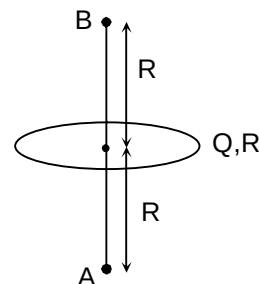
- (A) $e^{-\frac{\rho g H}{2c}}$
 (B) $\frac{\rho g H}{2c}$
 (C) $e^{-\frac{\rho g H}{c}}$
 (D) $\frac{\rho g H}{c}$

2. A uniform disc is held in equilibrium on a rough horizontal surface (coefficient of friction = μ) by applying a force as shown. Then the maximum value of F for this to be possible is (weight of disc is W).



- (A) zero
 (B) $\frac{2W}{\sqrt{3}}$
 (C) $\frac{\mu W}{1 + \sqrt{3}\mu}$
 (D) $\frac{2\mu W}{1 + \sqrt{3}\mu}$

3. A system of two charge particle A and B having charges $+q$ and $-q$ and mass m and $2m$ respectively and connected with an ideal massless inextensible string which lies along the axis of a uniform charged fixed non-conducting ring as shown. If gravity is neglected and ignore the force between A and B, find the tension in the string at this instant.



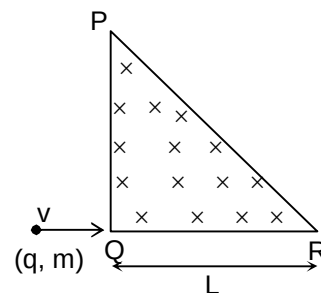
- (A) zero
 (B) $\frac{kqQ}{3\sqrt{2}R^2}$
 (C) $\frac{kqQ}{2\sqrt{2}R^2}$
 (D) $\frac{kqQ}{6\sqrt{2}R^2}$

4. A body takes 5 minutes to cool down from 50° to 40°C in surrounding of temperature 20°C . Then the time taken by the same body to cool down from 40°C to 30°C in a surrounding of temperature 10°C is (Assuming Newton's law of cooling is valid)

(A) 10 minutes
(B) 20 minutes
(C) 40 minutes
(D) 80 minutes

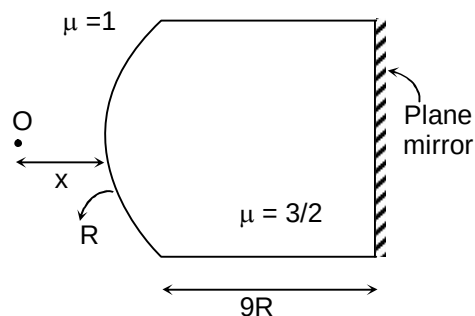
5. A positively charged particle having charge q and mass m enters in isosceles right angle triangular region having uniform magnetic field B as shown in gravity free space. The maximum value of v so that particle must comes out through the region from side PQ is

(A) $\frac{BqL}{\sqrt{2}m}$
(B) $\frac{(\sqrt{2}-1)BqL}{m}$
(C) $\frac{BqL}{2\sqrt{2}m}$
(D) $\frac{(\sqrt{2}-1)BqL}{2m}$



6. If the final image of the object O coincides with it then the sum of all possible values of x is

(A) $5R$
(B) $4R$
(C) $3R$
(D) $2R$

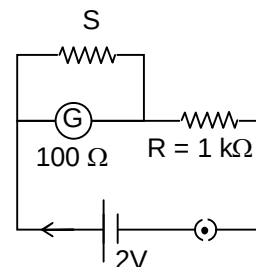


7. A tennis ball is thrown in downward direction with speed v from height h towards fixed horizontal surface with which it collides inelastically with coefficient of restitution e . The minimum value of v so that the ball will be able to reach at the point of projection is

(A) $\sqrt{2gh(1-e)}$
(B) $\sqrt{2gh(1-e^2)}$
(C) $\sqrt{2gh\left(\frac{1}{e}-1\right)}$
(D) $\sqrt{2gh\left(\frac{1}{e^2}-1\right)}$

8. A galvanometer is connected as shown in figure. What should be the resistance connected to it in parallel so that its deflection is reduced to half

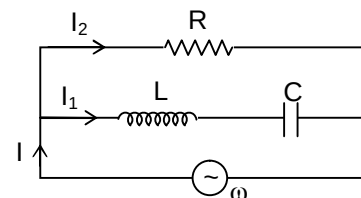
(A) 90Ω
 (B) 91Ω
 (C) 99Ω
 (D) 100Ω



9. A particle is moved along the curve $x^2 + y^2 = 2xy$ from $(0, 0)$ to $(1, 1)$ by some forces. If one among them given as $\vec{F} = y^2\hat{i} + 2xy\hat{j}$. Then the work done by the given force is

(A) zero
 (B) -1 Joule
 (C) 1 Joule
 (D) 2 Joule

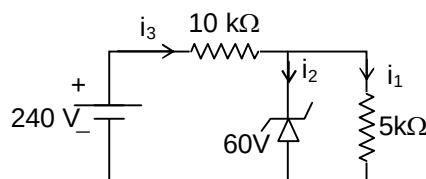
10. Which of the following statement is INCORRECT for the given circuit.



(A) For $\omega > \frac{1}{\sqrt{LC}}$, I_1 leads I_2
 (B) For $\omega > \frac{1}{\sqrt{LC}}$, phase difference between I_1 and I_2 is 90°
 (C) for $\omega < \frac{1}{\sqrt{LC}}$, phase difference between I_1 and I_2 is 90°
 (D) for $\omega = \frac{1}{\sqrt{LC}}$, is not permissible for the circuit.

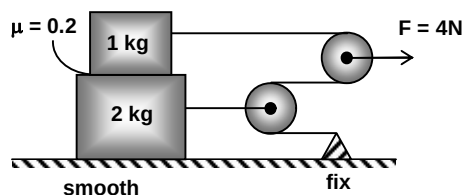
11. The correct value of current (i_1, i_2, i_3) in the circuit shown in the figure is

(A) $i_1 = 12 \text{ mA}$, $i_2 = 6 \text{ mA}$, $i_3 = 18 \text{ mA}$
 (B) $i_1 = 12 \text{ mA}$, $i_2 = 4 \text{ mA}$, $i_3 = 16 \text{ mA}$
 (C) $i_1 = 16 \text{ mA}$, $i_2 = 0 \text{ mA}$, $i_3 = 16 \text{ mA}$
 (D) $i_1 = 16 \text{ mA}$, $i_2 = 8 \text{ mA}$, $i_3 = 8 \text{ Ma}$

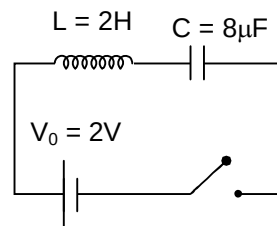


12. All the pulleys and strings are ideal and massless and the system shown is at rest. Now a force $F = 4\text{N}$ is applied on the pulley shown. Then the frictional force acting between the blocks is

(A) 2 N
 (B) 1.96 N
 (C) 3 N
 (D) 0 N

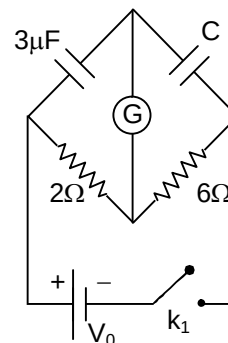


13. The speed of sound in hydrogen gas is v_0 at temperature T_0 . Then the speed of sound in the mixture of helium and oxygen gas having same number of moles at temperature T_0 is
- (A) $\frac{v_0}{3}$
- (B) $\sqrt{\frac{3}{14}}v_0$
- (C) $\sqrt{\frac{5}{42}}v_0$
- (D) $\sqrt{\frac{5}{21}}v_0$
14. An ideal gas is expanding such that $PT^2 = \text{constant}$. The coefficient of volume expansion of the gas is
- (A) $\frac{1}{T}$
- (B) $\frac{2}{T}$
- (C) $\frac{3}{T}$
- (D) $\frac{4}{T}$
15. A small smooth metal sphere of radius r and density ρ falls from rest in a viscous liquid of density σ and coefficient of viscosity η . Due to friction, heat is produced. The rate of production of heat when the sphere has acquired the terminal velocity is proportional to
- (A) r^2
- (B) r^3
- (C) r^4
- (D) r^5
16. Magnifying power of an astronomical telescope in near point adjustment is 110% of its magnifying power in normal adjustment. Then the focal length of objective when the length of the telescope in normal adjustment is 10 cm. (Take least distance of clear vision = 25 cm)
- (A) 2 cm
- (B) 2.5 cm
- (C) 7.5 cm
- (D) 8 cm
17. Initially the capacitor is uncharged and now switch is closed. Then the maximum current through the circuit is
- (A) 1 mA
- (B) 4 mA
- (C) 2 mA
- (D) 8 mA



18. If key is closed in the circuit shown in the figure and the galvanometer does not give any deflection at any time, then the value of C is

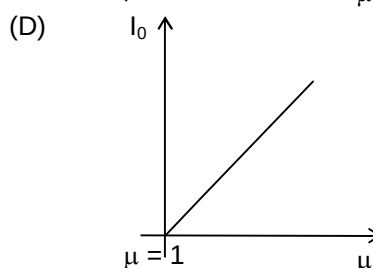
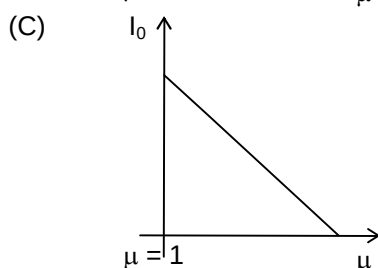
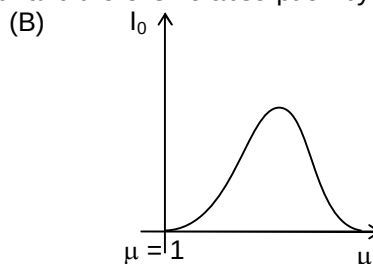
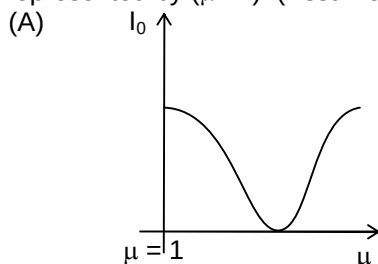
(A) $3 \mu\text{F}$
 (B) $9 \mu\text{F}$
 (C) $4 \mu\text{F}$
 (D) $1 \mu\text{F}$



19. A parallel beam of light having intensity $I = 10 \text{ Wm}^{-2}$ is continuously falling on emitter plate in the photoelectric effect setup at angle of 37° with the normal. If area of plate is 1 m^2 and wavelength of light is 621 nm and one photon is capable to eject one electron. Then saturation current in the circuit is ($\lambda_{\text{th}} > 621 \text{ nm}$)

(A) 3 A
 (B) 4 A
 (C) 1.8 A
 (D) 3.2 A

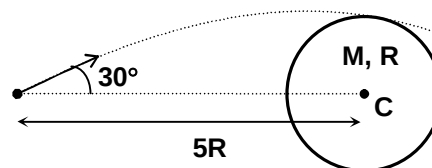
20. In a YDSE experiment, if a slab whose refractive index can be varied is placed in front of one of the slits then the variation of resultant intensity at mid-point of screen with ' μ ' will be best represented by ($\mu \geq 1$). (Assume slits of equal width and there is no absorption by slab)



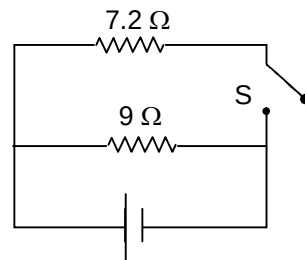
SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXX.XX).

21. A particle is thrown towards a stationary planet with speed $\sqrt{\frac{4kGM}{105R}}$ so that it grazes the planet as shown. Find k .

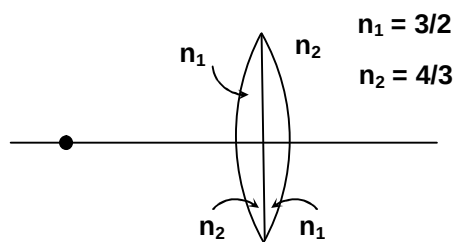


22. Two source resonance are observed at length 53.8 cm and 75.8 cm in a resonance column tube. Then the radius of the tube (in cm) is R. Find the value of R.
23. A projectile is thrown from ground towards a large vertical wall which is at a distance of 8 m from the point of projection. If launching speed of projectile from ground is $10\sqrt{2}$ m/s. If projectile hits the wall at a height h from ground. Then maximum possible value of h is(in meter) (Take $g = 10 \text{ m/s}^2$)
24. k_α wavelength emitted by an atom of atomic number $z = 11$ is λ . Then the atomic no. of an atom that emits k_α radiations of wavelength 4λ .
25. If power directed by the cell to the circle is same either switch is closed or open. Then the internal resistance of cell is $r \Omega$. Find the value of r.



26. If 3^{rd} excitation potential for a hypothetical hydrogen like atom is 15 volt. Then the value of 2^{nd} ionization energy for this hypothetical atom is N eV. Find the value of N.
27. Heat is given to an ideal gas (closed system) so that pressure and volume are increased by 10% and 20% respectively. Then percentage increase in temperature is N. Find the value of N.
28. Two Polaroids are placed on x-axis such that their planes are parallel to y-z plane and pass axis of these polaroids are making an angle of 45° with x-axis and also perpendicular to each other. If an unpolarised light of intensity I_0 propagating along x-axis having intensity after passes through the system of two polaroids is $\frac{KI_0}{16}$. Then find K.

29. The total number of images formed by the lens shown if the object is placed at O.



30. Assume that an electron and a positron pair is formed from a gamma ray photon having energy 3.0 MeV. Find the total kinetic energy (in MeV) of the system of electron and positron that is formed. (Given rest mass energy of electron is 0.51 MeV)

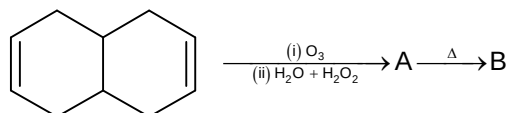
Chemistry

PART – B

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

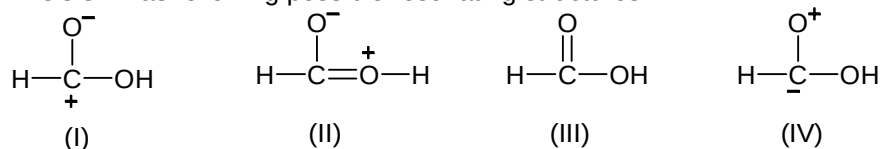
31.



Identify product B?

- (A)
- (B)
- (C)
- (D)

32. HCOOH has following possible resonating structures



Decreasing order of stability is:

- (A) I > II > III > IV
- (B) III > IV > I > II
- (C) III > II > I > IV
- (D) II > III > I > IV

 33. Identify which of the following reaction has negative $\Delta_r S$ (change in entropy)

- (A) $\xrightarrow{\Delta} \text{CH}_3 - \text{CH} = \text{CH}_2$
- (B) $\text{CH}_3\text{COO}^- (\text{aq}) + \text{H}_3\text{O}^+ (\text{aq}) \longrightarrow \text{CH}_3\text{COOH} + \text{H}_2\text{O}$
- (C) $\text{S}(\text{s}) \xrightarrow{\Delta} \text{S}(\text{g})$
- (D) $\text{H}_2 + \text{CH}_2 = \text{CH}_2 \longrightarrow \text{CH}_3 - \text{CH}_3$

34. Correct sequence of filling of electrons for 6s, 6p, 5d, 4f will be

- (A) $6s \rightarrow 6p \rightarrow 5d \rightarrow 4f$
 (B) $6s \rightarrow 5d \rightarrow 4f \rightarrow 6p$
 (C) $5d \rightarrow 6s \rightarrow 6p \rightarrow 5d$
 (D) $6s \rightarrow 4f \rightarrow 5d \rightarrow 6p$

35. For following electron gain enthalpies

- (i) $O(g) + e \longrightarrow O^-(g) \quad \Delta_{eg}H_1 = -141 \text{ kJ/mol}$
 (ii) $O^-(g) + e \longrightarrow O^{2-}(g) \quad \Delta_{eg}H_2$
 (iii) $O(g) + 2e \longrightarrow O^{2-}(g) \quad \Delta_{eg}H = \Delta_{eg}H_1 + \Delta_{eg}H_2$

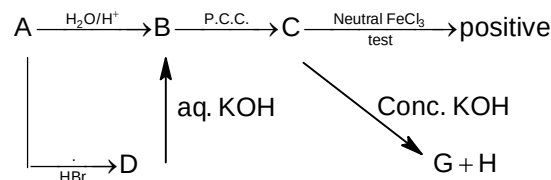
Identify correct statements:

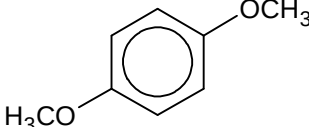
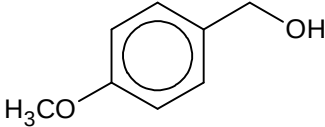
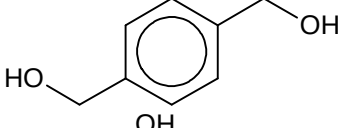
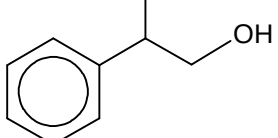
- (A) $\Delta_{eg}H_2$ is more negative than $\Delta_{eg}H_1$
 (B) $\Delta_{eg}H$ for reaction (iii) will be negative
 (C) $\Delta_{eg}H_2$ will be positive
 (D) $\Delta_{eg}H_1$ and $\Delta_{eg}H_2$ have almost same value

36. The shape of ICl_4^+ , ICl_4^- , and ICl_5 are respectively

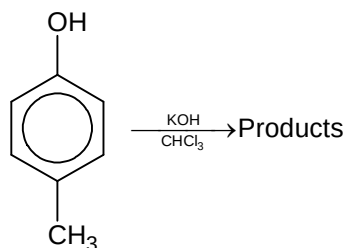
- (A) Trigonal bipyramidal, Tetrahedral and Tetrahedral
 (B) See saw, Square planer and Square pyramidal
 (C) See saw, Tetrahedral and Octahedral
 (D) Trigonal bipyramidal, Trigonal bipyramidal and Octahedral

37. Identify compound A ($C_8H_8O_2$) which gives following reaction



- (A) 
 (B) 
 (C) 
 (D) 

38.



Which of the following product is not possible?

- (A)
- (B)
- (C)
- (D) Both (A) and (B)

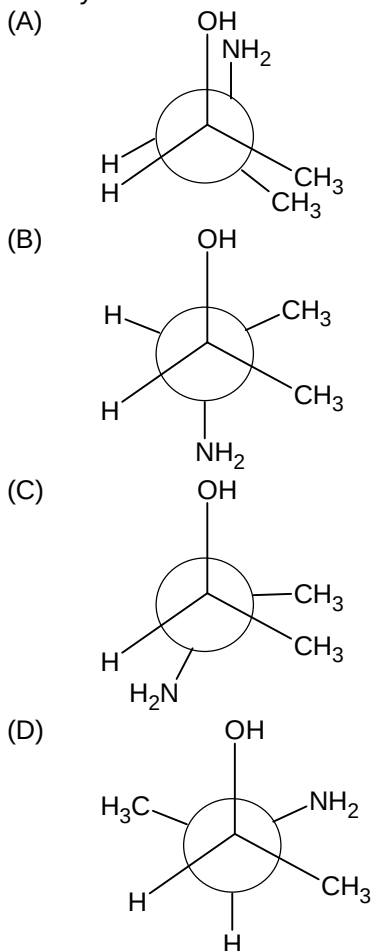
39. Identify wrong statement

- (A) Rickets and osteomalacia disease cause by deficiency of Vitamin-D.
- (B) Vitamin-A, D,E and K are stored in liver and adipose (fat storing) tissues.
- (C) Cellulose is a branched chain polysaccharide composed both α -D-Glucose and β -D-Glucose
- (D) During denaturation secondary and tertiary structures are destroyed but primary structure remains intact.

 40. 1.5 moles of gas A and y moles of gas B exert a pressure of 50 Pa and volume of container is 10 m³ at 200 K. y is ($R = \text{JK}^{-1} \text{mol}^{-1}$)

- (A) $\left(\frac{1-6R}{4R} \right)$
- (B) $\left(\frac{1-R}{4R} \right)$
- (C) $\frac{4-R}{2R}$
- (D) $\frac{R}{1-4R}$

41. Identify which of conformer of 3-amino-2-butanol is most stable?



42. Equilibrium constant for the mutarotation of glucose, α - D - glucose $\rightleftharpoons$ β -D-glucose is 1.4 percent of α - form remains under equilibrium is.....

- (A) 30
(B) 45
(C) 34.5
(D) 41.67

43. Identify correct statement

- (A) Powder of kernel of tamarind seeds has been found to an effective material to make municipal and industrial water waste clean.
(B) Maximum limit of nitrate in drinking water is 100 ppm.
(C) Lead can not damage kidney, liver and reproductive systems.
(D) Clean water would have BOD value more than 17 ppm.

44. Total number of possible isomers for $[\text{Pt}(\text{NH}_3)_4\text{Br}_2]\text{Cl}_2$?

- (A) 5
(B) 7
(C) 6
(D) 4

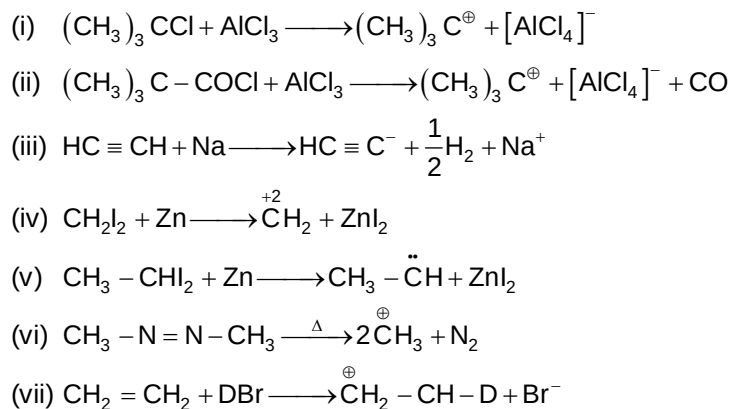
45. Which of the following is not rubber?
 (A) Neoprene
 (B) Buna-N
 (C) Vulcanised rubber
 (D) Teflon
46. Value of ΔG in the process of melting ice at -10°C is
 (A) equal to zero
 (B) less than zero
 (C) greater than zero
 (D) none of these
47. Which of the following reaction occur at cathode when electrolysis of $\text{NiCl}_2(\text{aq})$ take place at Pt electrode. ($E^\circ_{\text{Ni}^{+2}/\text{Ni}} = -0.25\text{ V}$)
 (A) $\text{Ni}^{+2} + 2\text{e} \longrightarrow \text{Ni(s)}$
 (B) $\text{Pt}^{+2} + 2\text{e} \longrightarrow \text{Pt(s)}$
 (C) $2\text{H}_2\text{O} + 2\text{e} \longrightarrow \text{H}_2 + 2\text{OH}^-$
 (D) $2\text{H}^+ + 2\text{e} \longrightarrow \text{H}_2$
48. Which of the following reaction is incorrect?
 (A) $\text{NH}_4\text{NO}_3 \xrightarrow{\Delta} \text{N}_2\text{O} + \text{H}_2\text{O}$
 (B) $\text{NH}_4\text{NO}_2 \xrightarrow{\Delta} \text{N}_2 + \text{H}_2\text{O}$
 (C) $\text{Sn} + \text{HNO}_3(\text{dil.}) \longrightarrow \text{Sn}(\text{NO}_3)_2 + \text{NO}_2 + \text{H}_2\text{O}$
 (D) $\text{Fe} + \text{HNO}_3(\text{very dil.}) \longrightarrow \text{Fe}(\text{NO}_3)_2 + \text{NH}_4\text{NO}_3 + \text{H}_2\text{O}$
49. Total number of S – O – S linkages in γ form of SO_3 are
 (A) 2
 (B) 3
 (C) 4
 (D) 5
50. In which of the following ore carbonate is present?
 (A) Limonite
 (B) Zincite
 (C) Diaspore
 (D) Siderite

SECTION – B (Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXX.XX).

51. 10 gm carboxylic acid ($\text{C}_x\text{H}_y\text{O}_z$) with molecular mass 118 gm/mole, neutralizes 169.5 ml of 1 M KOH. When 1 mole of carboxylic acid is heated 1 mole of H_2O release without loss of CO_2 . Total number of carbon atoms and hydrogen atoms in the carboxylic acid are x and y respectively. Find y/x?

52. "a" and "b" are total number of correct and incorrect reactions respectively. Find a/b?

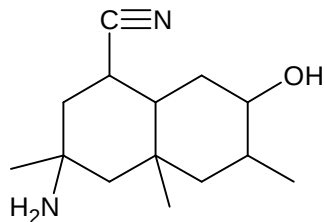


53. 1.5 gm of O_2 and some NaCl is formed by heating sample of NaClO_3 . The residue is reacted with excess of AgNO_3 and form AgCl. Mass of AgCl (in gram) will be (answer in two decimal places)

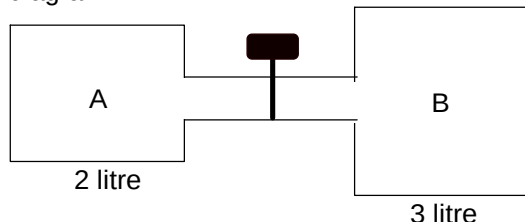
54. x are total number of correct statements

- (i) Dihydrate of calcium sulphate is known as gypsum.
 (ii) None alkaline earth metal hydrides are ionic in nature.
 (iii) Phenol gives paraquinol with $\text{S}_2\text{O}_8^{2-} / \text{OH}^{-}$ as major product.
 (iv) Phenol can give electrophilic aromatic substitution in absence of Lewis acid (AlCl_3).
 (v) Solvay process can not be used to prepare K_2CO_3 due to precipitation of KHCO_3 .
 (vi) Salicylic acid can give white/yellow solution with $\text{Br}_2 / \text{CS}_2$. Find $\left(\frac{x+5}{4}\right)$?

55. x are the total number of chiral carbon atoms and y are total number of hydrogen atoms which are attached to chiral carbon atoms



56. Chamber A and B are filled with 2 atm He and 4 atm H_2 respectively. Both chamber are separated by small tube (negligible volume) with stopcock at 300 K temperature as given in diagram



After opening stopcock find out total pressure (atm) in chamber A.

57. Solution of 0.05 mole of sugar in 500 gm of water shows 0.279°C freezing point. Cane sugar solution has inversion as follows
 Cane sugar + $\text{H}_2\text{O}(\ell) \longrightarrow \text{Glucose} + \text{Fructose}$
 Find out mass of cane sugar left without inversion. (m.wt. of cane sugar = 342 gm/mole)
58. Surfactant solution of 100 ml 50 mM solution forms a monolayer covering 67.5 cm^2 on a polar substrate. If polar head is approximate as a cube. What is its edge length in Pm? ($N_0 = 6 \times 10^{23}$)
59. A gas equation is $P(V - nb) = nRT$ for a state, where $V_m = 5b$, for any pressure and temperature. Compressibility factor of gas for this condition is.....(b = van der Waal's constant, R = Universal gas constant, V_m = molar volume).
60. Enthalpy change (in kcal) for the reaction $\text{XeF}_2 \longrightarrow \text{Xe}^+(\text{g}) + \text{F}^-(\text{g}) + \text{F}(\text{g})$ is 'x'. Find out $\frac{x}{100}$?
 (Average bond enthalpy of Xe – F = 34 kcal/mol, 1^{st} ionization enthalpy of Xe (g) = 279 kcal/mol, Electron gain enthalpy of F(g) = - 85 kcal/mol).

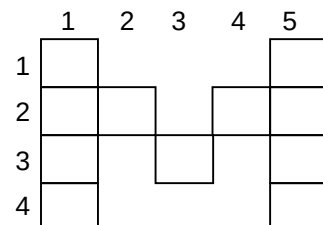
Mathematics

PART – C

SECTION – A (One Options Correct Type)

This section contains **20 multiple choice questions**. Each question has **four choices** (A), (B), (C) and (D), out of which **ONLY ONE** option is correct.

61. Number of ways in which first 11 natural numbers can be arranged in the given figure such that $a_{ij} > a_{kj}$ whenever $i > k \forall j$ is (a_{ij} represents element in i^{th} row and j^{th} column)
- (A) 69300
(B) 11550
(C) 34650
(D) None of these



62. Let $f(x) = 2x^2 + 4x + 9$ and $g(x) = f(\cos^2 x - \sin x)$, then number of points of extrema in $x \in (0, 2\pi)$ is
- (A) 3
(B) 4
(C) 5
(D) 6
63. Let $f(x) = \begin{cases} \sqrt{x-2k} & ; 2k \leq x < 2k+1 \\ \sqrt{2k+2-x} & ; 2k+1 \leq x < 2k+2 \end{cases}$, where $k \in \mathbb{I}$. Consider the following statements:
- (1) $f(x)$ is even function
(2) $f(x)$ is periodic function
(3) range of $f(x)$ is $[0, \infty)$
(4) $f(x) = f(2-x) \forall x \in \mathbb{R}$
- The correct statements are
- (A) 1, 2, 3
(B) 1, 2, 4
(C) 2, 3, 4
(D) 1, 3, 4
64. The solution of the differential equation
- $$dy \left(x^2y - \frac{x^4y^3}{3} + \frac{x^6y^5}{5} - \frac{x^8y^7}{7} + \dots \right) + dx \left(xy^2 - \frac{x^3y^4}{3} + \frac{x^5y^6}{5} - \frac{x^7y^8}{7} + \dots \right) = 0$$
- (A) $xy \tan^{-1}(xy) + \ln(1+x^2y^2) = c$
(B) $xy \tan^{-1}(xy) - \ln(1+x^2y^2) = c$
(C) $xy \tan^{-1}(xy) + \frac{1}{2} \ln(1+x^2y^2) = c$
(D) $xy \tan^{-1}(xy) - \frac{1}{2} \ln(1+x^2y^2) = c$
65. Variance of set of 100 observation given by 3, 5, 7, 9, is
- (A) 9999
(B) 6666
(C) 3333
(D) 1111

66. Logical statement $(p \leftrightarrow q) \leftrightarrow (q \leftrightarrow r)$ is equivalent to
 (A) $(p \wedge r) \vee (\neg p \wedge \neg q \wedge \neg r)$
 (B) $(p \wedge r) \vee (\neg p \wedge \neg q) \vee (\neg q \wedge \neg r)$
 (C) $(p \wedge r) \vee (\neg p \wedge q \wedge \neg r) \vee (\neg p \wedge \neg q \wedge \neg r)$
 (D) $(p \wedge r) \vee (\neg p \wedge \neg q \wedge r) \vee (p \wedge \neg r \wedge q)$
67. Let $R = \{(a, b) \mid a - b \text{ is irrational ; } a, b \text{ are real numbers}\}$, then relation R is
 (A) reflexive and symmetric relation
 (B) transitive and symmetric
 (C) symmetric relation
 (D) equivalence relation
68. There is a circular park of radius 100 m having two trees of equal height h on its circumference. A person is standing on the circumference of the park such that the angle of elevation of top of both the trees is equal to 30° . The person starts running at a speed of 10 m/s along the periphery and reaches one of the tree in 5 seconds, then h equals (in metres)
 (A) $\frac{200}{\sqrt{3}} \sin\left(\frac{1}{4}\right)$
 (B) $\frac{200}{\sqrt{3}} \cos\left(\frac{1}{4}\right)$
 (C) $\frac{200}{\sqrt{3}} \sin\left(\frac{1}{2}\right)$
 (D) $\frac{200}{\sqrt{3}} \cos\left(\frac{1}{2}\right)$
69. If $f(x) = xf^2(x) + x^2 \forall x$, then $\int \frac{f^2(x) - 2xf(x) + 2x + 1}{(1 - 2xf(x))f(x)(f(x)(1 - x) + 2x + 1)} dx$ equals
 (A) $\frac{-1}{x - f(x)} + c$
 (B) $\frac{-1}{x + f(x)} + c$
 (C) $\frac{-1}{x^2 - f(x)} + c$
 (D) $\frac{-1}{x^2 + f(x)} + c$
70. Let $2x^2 + 2y^2 - 5xy = 0$, $y = mx + c$ form a triangle such that reflection of any point (x, y) , lying inside the triangle, in the line $y = m'x$ also lies inside the triangle, then minimum possible value of $m - m'$ equals
 (A) -2
 (B) 0
 (C) 2
 (D) None of these

71. Given a point P from which tangents T_1 and T_2 are drawn to a unit circle C_i such that the chord of contact so formed subtends $\frac{2\pi}{3}$ at the centre of circle. Now, circle C_{i+1} is drawn which touches T_1 , T_2 and C_i externally ($i = 1, 2, 3, \dots$). If r_i represents radius of circle C_i and $r_{i+1} < r_i \forall i \in \mathbb{N}$, then $\lim_{n \rightarrow \infty} \sum_{i=1}^n r_i$ equals
- (A) $\frac{20}{7}$
 (B) $\frac{31}{9}$
 (C) $\frac{5}{3}$
 (D) $\frac{3}{2}$
72. Let a mirror be placed along the parabola $y^2 = 4x$. A ray of light parallel to x-axis is incident on the mirror at point P and after reflection is incident on the parabola again at point Q. If reflected ray at Q is same as incident ray at Q, then coordinates of P are
- (A) $(2, 2\sqrt{2})$
 (B) $(3, 2\sqrt{3})$
 (C) $(4, 4)$
 (D) P does not exist
73. ${}^{674}C_0 {}^{2022}C_{674} - {}^{674}C_1 {}^{2020}C_{674} + {}^{674}C_2 {}^{2018}C_{674} - \dots + {}^{674}C_{674} {}^{674}C_{674}$ equals
- (A) 2^{674}
 (B) 2^{1348}
 (C) 2^{2022}
 (D) None of these
74. Let A be set of all 2×2 symmetric nilpotent real matrices and B be set of all 2×2 skew symmetric nilpotent real matrices, then $n(A \cup B)$ equals
- (A) 0
 (B) 1
 (C) 4
 (D) None of these
75. Number of solutions of the equation $\sqrt{3} \tan x + 3\sqrt{3} \cot^3 x - 4 \sin^2\left(\frac{\pi}{6} + x\right) = 0$ in $x \in [0, 2022\pi]$ is equal to
- (A) 2022
 (B) 4044
 (C) 8088
 (D) None of these

76. A given circle has chord AB of length 4 units, chords BC of length 5 units and chord AC of length 6 units, then area of the circle is
- (A) $\frac{16\pi}{7}$
 (B) $\frac{32\pi}{7}$
 (C) $\frac{64\pi}{7}$
 (D) $\frac{192\pi}{7}$
77. Number of values of x satisfying the system of equation $\sin^{-1}\left(\frac{2x}{1+x^2}\right) = \cos^{-1}\left(\frac{1-x^2}{1+x^2}\right) = 2\cos^{-1}x$ is
- (A) 0
 (B) 1
 (C) 2
 (D) infinite
78. Let $P_1 : x + y + z = 0$, $P_2 : x + y + 2z = 1$, $P_3 : ax + by + 3z = 1$. If P_1, P_2, P_3 have infinite points of intersection, then the value of $a + b$ such that distance of P_3 from $x - 1 = \frac{y-1}{2} = \frac{z-1}{3}$ is maximum is
- (A) 3
 (B) -3
 (C) 6
 (D) -6
79. If all values of $x \in (a, b)$ satisfy the inequality $\tan x \tan 3x + 1 < 0$, $x \in \left(0, \frac{\pi}{2}\right)$, then maximum possible value of $(b - a)$ equals
- (A) $\frac{\pi}{12}$
 (B) $\frac{\pi}{6}$
 (C) $\frac{\pi}{4}$
 (D) $\frac{\pi}{3}$
80. Let $\vec{a}, \vec{b}$ be two non-zero vectors representing points A and B respectively in space. If A and B lie on plane P and any other vector $\vec{r}$ in plane P can be written as $\vec{r} = \alpha\vec{a} + \beta\vec{b}$ ($\alpha, \beta \in \mathbb{R}$), then
- (A) origin cannot lie on P
 (B) origin must lie on line AB
 (C) origin can lie anywhere on P except on line AB
 (D) nothing can be said about location of origin based on given information

SECTION – B
(Numerical Answer Type)

This section contains **10** questions. The answer to each question is a **NUMERICAL VALUE**. For each question, enter the correct numerical value (in decimal notation, truncated/rounded-off to the **second decimal place**; e.g. XXXX.XX).

81. If letters of the word 'IITJEE' are **rearranged**, then probability that J appears before T is $\frac{p}{q}$ (where p and q are coprime), then the value of p + q is
82. Let $f(x) = \max\{2^{\cos x}, |\sin x|, |\cos x|\}$ for $x \in (0, 2\pi)$, then the number of critical points of f(x) is
83. Let $f(x) = [\sin x + [\cos x + [\tan x + [\cot x + [\operatorname{cosec} x + [\sec x]]]]]]$ (where [.] denotes greatest integer function)
 $A = \left\{ y \mid y = f(x); x \in \left(0, \frac{\pi}{4}\right) \right\}$; $B = \left\{ y \mid y = f(x); x \in \left(\frac{\pi}{2}, \frac{3\pi}{4}\right) \right\}$ and I be set of all integers, then $n((I - A) \cap (I - B))$ is equal to
84. $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{2022 - 2023 \operatorname{cosec}^2 x}{\sin^{2022} x} dx$ equals $-\frac{2^p(3^q - 1)}{3^r}$, then p + q + r is equal to
85. If $f(x + f(y)) + f(x - f(y)) = f(x) \forall x, y \in \mathbb{R}$ and (0, 2) lies on $y = f(x)$, then $\int_0^{12} f(x) dx = \lambda \int_{-24}^{24} f(x - 36) dx$ where λ equals
86. Number of ways to distribute 5000 similar objects to 100 people $P_1, P_2, \dots, P_{100}$ such that P_i gets atleast (i - 1) objects equals nC_r , then maximum value of n + r equals
87. Let L_1 : tangent to $y = \sin x$ at $x = 0$, L_2 : tangent to $y = \sin x$ at $x = \frac{\pi}{2}$, L_3 : tangent to $y = \sin x$ at $x = \pi$, then area bounded by L_1, L_2, L_3 and $y = 0$ equals
88. Value of r such that the system of equations $|z| = r, |z - 6| = r, |z - 3 - 3\sqrt{3}i| = r$ have a unique solution is
89. Let $m = \lim_{x \rightarrow 0^+} \left[\frac{e^{\operatorname{sgn} x} - 1}{\operatorname{sgn} x} \right]$ and $n = \lim_{x \rightarrow 0^-} \left[\frac{e^{\operatorname{sgn} x} - 1}{\operatorname{sgn} x} \right]$ (where [.] denotes greatest integer function and $\operatorname{sgn}(x)$ represents signum function), then m - n equals
90. $\frac{1}{1.2} + \frac{1}{3.4} + \frac{1}{5.6} + \frac{1}{7.8} + \dots$ equals